

JEE MAIN REVISION MASTER: GEOMETRICAL OPTICS

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 268 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Reflection at Spherical Mirrors & Mirror Formula	122 PYQs	Mirrors
Very High	Refraction at Interfaces, Snell's Law & TIR	110 PYQs	Refraction
Very High	Lens Maker's Formula & Optical Instruments	70 PYQs	Lenses & Instruments
Very High	Prism Dispersion & Minimum Deviation	39 PYQs	Prisms

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Reflection at Spherical Mirrors & Mirror Formula	$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$, $\text{quad } m = -\frac{v}{u} = \frac{f}{f - u}$, $\text{quad } f = \frac{R}{2}$
Refraction at Interfaces, Snell's Law & TIR	$n_1 \sin i = n_2 \sin r$, $\text{quad } \sin \theta_c = \frac{n_2}{n_1}$; ($n_1 > n_2$), $\text{quad } d' = \frac{d}{\mu}$, $\text{quad } \text{Shift} = t \left(1 - \frac{1}{\mu} \right)$
Lens Maker's Formula & Optical Instruments	$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$, $\text{quad } P = \frac{1}{f}$, $\text{quad } M_{\text{telescope}} = \frac{f_o}{f_e}$, $\text{quad } M_{\text{microscope}} = \frac{L}{f_o} \frac{D}{f_e}$
Prism Dispersion & Minimum Deviation	$\delta = i + e - A$, $\text{quad } \mu = \frac{\sin \left(\frac{A + \delta_m}{2} \right)}{\sin(A/2)}$, $\text{quad } \delta \approx (\mu - 1)A$; (thin prism)

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Reflection at Spherical Mirrors & Mirror Formula (Very High Yield — 122 PYQs)

Core Principle: Image formation by concave and convex mirrors, Cartesian sign convention, and linear/longitudinal magnification.

■ **Caution / JEE Pitfall:** Concave mirror has negative focal length ($f < 0$); convex mirror has positive focal length ($f > 0$).

■ **Speed Trick:** Virtual image in a concave mirror is ALWAYS magnified ($|m| > 1$), while in a convex mirror it is ALWAYS diminished ($|m| < 1$).

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "A spherical interface lens of radius R separates two media of refractive indices 1 and 1.4 respectively as shown in the figure below. A po..." [Ans: A]
- [2026 Morning Shift] Q#1 (MCQ): "A compound microscope is designed with two symmetric biconvex lenses. The objective lens is cut vertically, creating two identical plano-con..." [Ans: A]
- [2026 Evening Shift] Q#1 (MCQ): "In parallax method for the determination of focal length of a concave mirror, the object should always be placed:..." [Ans: C]

3.2 Refraction at Interfaces, Snell's Law & TIR (Very High Yield — 110 PYQs)

Core Principle: Snell's law, apparent depth shift in transparent media, and total internal reflection condition ($i \geq \theta_c$ from denser to rarer).

■ **Caution / JEE Pitfall:** Total internal reflection can ONLY occur when light travels from an optically DENSER medium to a RARER medium.

■ **Speed Trick:** Area of circle through which light emerges from fish/source at depth h : $A = (\pi h^2 / \mu^2 - 1)$.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "A spherical interface lens of radius R separates two media of refractive indices 1 and 1.4 respectively as shown in the figure below. A po..." [Ans: A]
- [2026 Morning Shift] Q#1 (MCQ): "An unpolarised light is incident at an interface of two dielectric media having refractive indices of 2 (incident medium) and $2\sqrt{3}$ (m..." [Ans: B]
- [2026 Evening Shift] Q#1 (MCQ): "A prism of angle 75° and refractive index $\sqrt{3}$ is coated with thin film of refractive index 1.5 only at the back exit surface..." [Ans: N/A]

3.3 Lens Maker's Formula & Optical Instruments (Very High Yield — 70 PYQs)

Core Principle: Thin lens refraction, Lens Maker's equation, lens power in Dioptres ($P = 1/f$), and magnifying power of optical devices.

■ **Caution / JEE Pitfall:** When a lens is immersed in a liquid of refractive index $\mu_l > \mu_{\text{glass}}$, its converging/diverging nature REVERSES (convex acts concave).

■ **Speed Trick:** When a lens is cut vertically in half, each half retains the SAME focal length f . When cut horizontally, focal length remains f , but image brightness is halved.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "A compound microscope is designed with two symmetric biconvex lenses. The objective lens is cut vertically, creating two identical plano-con..." [Ans: A]
- [2026 Morning Shift] Q#1 (MCQ): "The magnitudes of power of a biconvex lens (refractive index 1.5) and that of a plano-concave lens (refractive index $=1.7$) are same. If t..." [Ans: B]
- [2026 Evening Shift] Q#2 (MCQ): "A thin biconvex lens is prepared from the glass ($\mu=1.5$) both curved surfaces of which have equal radii of 20 cm each. Left side surface ..." [Ans: A]

3.4 Prism Dispersion & Minimum Deviation (Very High Yield — 39 PYQs)

Core Principle: Ray refraction through triangular prisms, condition for minimum deviation ($i = e$, $r_1 = r_2 = A/2$), and angular dispersion.

■ **Caution / JEE Pitfall:** At minimum deviation, the refracted ray inside an equilateral prism passes strictly PARALLEL to the base.

■ **Speed Trick:** Condition for no emergent ray (always TIR on second face): $A > 2\theta_c$.

Representative Source PYQs:

- **[2026 Morning Shift] Q#1 (MCQ):** "For a thin symmetric prism made of glass (refractive index 1.5), the ratio of incident angle and minimum deviation will be _____." [Ans: B]
- **[2026 Morning Shift] Q#3 (MCQ):** "A thin prism with angle 5° of refractive index 1.72 is combined with another prism of refractive index 1.9 to produce dispersion wi..." [Ans: B]
- **[2026 Evening Shift] Q#4 (MCQ):** "Angle of minimum deviation is equal to the half of the angle of prism in an equilateral prism. The refractive index of the prism is _____." [Ans: C]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

A spherical interface lens of radius R separates two media of refractive indices 1 and 1.4 respectively as shown in the figure below. A point source is placed at a distance of $4R$ in front of spherical interface. The magnitude of the magnification of point source image is _____.

- (A) 1.66 [Correct]
- (B) 2.33
- (C) 2.66
- (D) 1.33

Correct Answer: A

Solution Analysis: When light travels from one medium to another through a curved surface, the positions of the object (u) and the image (v) are related by the formula: $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$ Where n_1 is the refractive index of the medium where the object is placed, n_2 is the refractive index of the medium where light enters, R is the radius of curvature. The refractive index of medium in which object is $n_1=1$, and that for the image $n_2=1.4$. The object distance is $u=-4R$ (as per sign convention), and radius is $+R$ (convex surface). $\frac{1.4}{v} - \frac{1}{-4R} = \frac{1.4 - 1}{R}$

Benchmark Problem 2: JEE Main 2026 (Online) 2nd April Morning Shift (Q#1)

For a thin symmetric prism made of glass (refractive index 1.5), the ratio of incident angle and minimum deviation will be _____.

- (A) 3 : 4
- (B) 3 : 2 [Correct]
- (C) 2 : 1
- (D) 1 : 2

Correct Answer: B

Solution Analysis: For a thin prism with a small prism angle (A), the angle of minimum deviation is given by, $\delta_m = (\mu - 1)A$. Where μ is the refractive index of the glass, $\mu = 1.5$: Putting the value of refractive index, $\delta_m = (1.5 - 1)A = 0.5A = (A / 2)$. At the position of minimum deviation, the angle of incidence (i) is related to the prism angle (A) and the minimum deviation (δ_m) by $i = \frac{A + \delta_m}{2}$. Substituting the value of $\delta_m = (A / 2)$, $i = (A + (A / 2)) / 2 = (3A / 2) / 2 = (3A / 4)$. So, the ratio of the incident angle (i) to the minimum deviation (δ_m) is $\frac{3A / 4}{A / 2} = \frac{3}{2}$.

Benchmark Problem 3: JEE Main 2026 (Online) 28th January Morning Shift (Q#1)

The magnitudes of power of a biconvex lens (refractive index 1.5) and that of a plano-concave lens (refractive index = 1.7) are same. If the curvature of planoconcave lens exactly matches with the curvature of back surface of the biconvex lens, then ratio of radius of curvature of front and back surface of the biconvex lens is _____

- (A) 2:5
- (B) 5:2 [Correct]
- (C) 12:5
- (D) 5:12

Correct Answer: B

Solution Analysis: If the focal length of a lens is f (in meters) then its power is given as $P = (1 / f)$... (i) The Lens Maker's formula is given as $(1 / f) = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$, where μ is the refractive index of lens with respect to the medium, R_1 and R_2 are the radii of curvatures (measured using sign convention). Using the Lens Maker's Formula and power formula : $P = (1 / f) = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$. Here, $\mu_1 =$ refractive index of biconvex lens, R_1 , R_2 are the radii of curvatures of front and back curvature of biconvex lens. The

Benchmark Problem 4: JEE Main 2026 (Online) 24th January Morning Shift (Q#1)

An unpolarised light is incident at an interface of two dielectric media having refractive indices of 2 (incident medium) and $2\sqrt{3}$ (medium) respectively. To satisfy the condition that reflected and refracted rays are perpendicular to each other, the angle of incidence is _____

- (A) 45°
- (B) 60° [Correct]
- (C) 10°
- (D) 30°

Correct Answer: B

Solution Analysis: Light hits the surface at an angle of incidence i . Part of the light bounces back. According to the Law of Reflection, the angle of reflection is equal to the angle of incidence (i). Part of the light enters the second medium. It bends at an angle of refraction r . The reflected and refracted rays are perpendicular to each other. This means the angle between the reflected ray and the refracted ray is 90° . The sum of angles on a straight line is 180° . $i + 90^\circ + r = 180^\circ \Rightarrow i + r = 90^\circ \Rightarrow r = 90^\circ - i$. Now using the fundamental law of re